Διόφαντος ὁ Ἀλεξανδρεύς

Diophantus of Alexandria and the Arithmetic Tradition

eul_aid: rqc

Period: *Ῥωμαϊκή* Roman
Dialect: *Μικταὶ διάλεκτοι* Mixed Dialects (frg)
Domain: *Μαθηματικά* Mathematics
Format: *Πραγματεία* Treatise

Diophantus of Alexandria was a Greek mathematician who lived during the Roman period, most likely in the 3rd century CE. Very little is known about his personal life. His age of 84 is recorded in a later mathematical riddle, but the exact dates of his birth and death are uncertain.

He is known for a single major work, the Arithmetica, a treatise on solving algebraic equations. Originally in 13 books, only 6 survive in Greek, with an additional 4 books preserved through a later Arabic translation. The work is a collection of over 100 problems. Many of these problems lead to what are now called Diophantine equations, where the goal is to find specific types of solutions, often positive rational numbers.

According to modern scholars, Diophantus's work represents a unique and original branch of Greek mathematics, separate from the dominant geometric tradition. His use of abbreviations for the unknown and its powers was a significant step toward the development of symbolic algebra. The Arithmetica had a major influence on later Islamic mathematicians and, after its translation into Latin in the 17th century, on European mathematics. Its most famous legacy is its connection to Pierre de Fermat, who wrote his celebrated "Last Theorem" in the margin of his copy of Diophantus's book.

Works (1)

- Ἀποσπάσματα περὶ Ποντικῶν Ἱστοριῶν · Fragments on Pontic Histories · 4 passages